RESEARCH PROPOSAL, INNOVATION FUNDING PROGRAM - P3 SCHEME, UNIVERSITAS INDONESIA

A. INNOVATION PROPOSAL SUMMARY			
A.1. Applicant Capacity			
1.1. List of Proposing Team Members	**Name** (complete with academic titles)	**Status**	**Job Description within the team**
	Chair: Dr. Palupi Lindiasari Samputra	UI Lecturer	Designing the concept of smart monitoring for community food security, and analyzing and developing the community food security index formula for the region
	Member 1: Dr. Stanislaus Riyanta, S.Si., M.Si	UI Lecturer	Analysis of the urban farming area environment, related to nutrient content (micronutrients), for mapping ecological resilience
	Member 2: *Saur MarthaAgustina,M.Si*	General	Research assistant and field coordinator
	Member 3: Eggan Nachson,M.Si.	General	Research Assistant for the design of the regional community food security system using AI and OSINT (Open Source Intelligence). Design of the dashboard, policy briefs, and monthly reports on the region's food security conditions
	Member 4: Lia Aprilia	UI Administrative Staff	Administration and finance
	Member 5: Yusril	Student	Member serving as a research assistant responsible for technical field duties (socializing the system with the community, assisting with the implementation of the research)
	Member 6: Hotman	UI Alumnus	UI alumnus serving as a liaison for the alumni and partner network, supporting program outreach and the sustainability of LUMBUNG's partnership with the community and external stakeholders
1.2. Number of Researchers Involved	UI Lecturers: 2 UI Administrative Staff: 1 Students: 1 UI Alumni/General: 3		
1.3. Number of Students Involved	Undergraduate: 3	Master's: 2 Doctoral: 2	

<table>
<tr><td colspan="4">A.2. Innovation Description</td></tr>
<tr><td>1.1. Innovation Proposal Title</td><td colspan="3">Smart Monitoring Food System Resilience - AI-Based Urban Farming Community in the North Jakarta Region.</td></tr>
<tr><td>1.2. Innovation Product Name</td><td colspan="2">LUMBUNG: Food System Resilience of the North Jakarta Region</td><td>Initial TKT (Technology Readiness Level): 7</td></tr>
<tr><td rowspan="2">1.3. Key Technology & Innovation Output</td><td colspan="2">Key Technology</td><td>Output</td></tr>
<tr><td colspan="2">AI, Data Coding, Dashboard Display</td><td>Product/Technology (Science, Technology, and Health)</td></tr>
<tr><td>1.4. Innovation Focus Topic</td><td colspan="2">Primary Innovation Focus:
Functional Food</td><td>Secondary Innovation Focus (if any):
Artificial Intelligence and Big Data</td></tr>
<tr><td rowspan="2">1.5. Partner</td><td rowspan="2">Partner Name:
Sudin KPKP (North Jakarta Sub-district Office of Food Security, Maritime Affairs and Agriculture)
Partner's Main Product:
Food System Resilience Monitoring Dashboard for North Jakarta</td><td colspan="2">Partner Type:: Government Institution</td></tr>
<tr><td colspan="2">Partner Category:: User Partner</td></tr>
<tr><td colspan="4">A.3. Funding Proposal</td></tr>
<tr><td colspan="2">1.1. Funds Proposed to UI</td><td colspan="2">Rp.200.000.000</td></tr>
<tr><td colspan="2">1.2 Partner Matching Funds (in-kind)</td><td>Rp.0</td><td>Partner Fund Percentage: 15 %
(of the funds proposed to UI)</td></tr>
</table>

B. SIGNIFICANCE AND BENEFIT OF THE INNOVATION (maximum 2 pages)

This section should be completed with an explanatory description of:

- *The problem statement regarding the appropriateness of the innovation*
- *Advantages/competitiveness of the innovation compared to existing ones*
- *The level of local raw material content in the innovation product*
- *Prior intellectual property (IP) held by the research team and the potential IP outputs as licensable key technology*
- *The urgency of the research for the partner: the significance and benefit of the innovation for the industrial and business sector and/or government institutions*
- *The target and market share of the resulting innovation product. The*

description may be in narrative form accompanied by images, diagrams, and/or tables.

Problem Statement

FAO data from 2023 indicates that 56% of urban areas were detected as having critical levels of food security. In Jakarta, with its limited land, particularly in North Jakarta, as many as 6 sub-districts have established several farming communities aimed at strengthening regional food security. The partner in this research is Sudin KPKP (North Jakarta), which faces limitations in monitoring and evaluating food security programs. This is due to a monitoring system that is manual, non-periodic, and based solely on field problem reports that are incidental in nature. Coordination between the UI research team, chaired by Dr. Palupi Lindiasari Samputra, and the Head of Sudin KPKP North Jakarta on 12 June 2026 revealed that the lack of real-time data access causes various decisions and program interventions to be delayed. This limitation is due to the absence of a well-planned, regularly scheduled monitoring and evaluation system. As a result, Sudin KPKP cannot directly measure the impact of interventions provided—such as food crop seeds, fisheries inputs, and other inputs (vitamins, fertilizer)—on the crop products produced by farming communities in the North Jakarta region. Currently, the available data is only quarterly and requires a long time to collect. Therefore, the existence of real-time technology is greatly expected by the partner to help the agency's work, from outreach, intervention, monitoring, and evaluation, to feedback for improving future programs.

Another problem hindering the achievement of the program's targets is that the mechanism for requesting agricultural, livestock, and fisheries assistance in the North Jakarta region is still limited to an informal process (via phone/message) without any ticketing or status tracking. Furthermore, there is no integrated system linking residents and the agency regarding food stock and prices within a single, jointly accessible platform. Finally, and most importantly, the government's response in detecting food crises is slow because data only becomes available after a manual survey is conducted. Manual surveys are not conducted regularly and are slow, as they require time to collect data.

The Urgency of Digital Adoption for Real-Time Monitoring for Sudin KPKP North Jakarta

The application of AI-based technology can serve as a solution for Sudin KPKP North Jakarta in real-time monitoring and evaluation efforts, as well as providing analysis of the region's food security and vulnerability conditions. This is highly important for accelerating government policy in intervening in food insecurity conditions. There are structural obstacles that need to be immediately addressed in the implementation of this dashboard, namely:

1) Food data is not centralized

2) Residents do not have an official channel
3) Assistance response is slow
4) There is no real-time food map
5) Community food prices are not monitored against the HET price

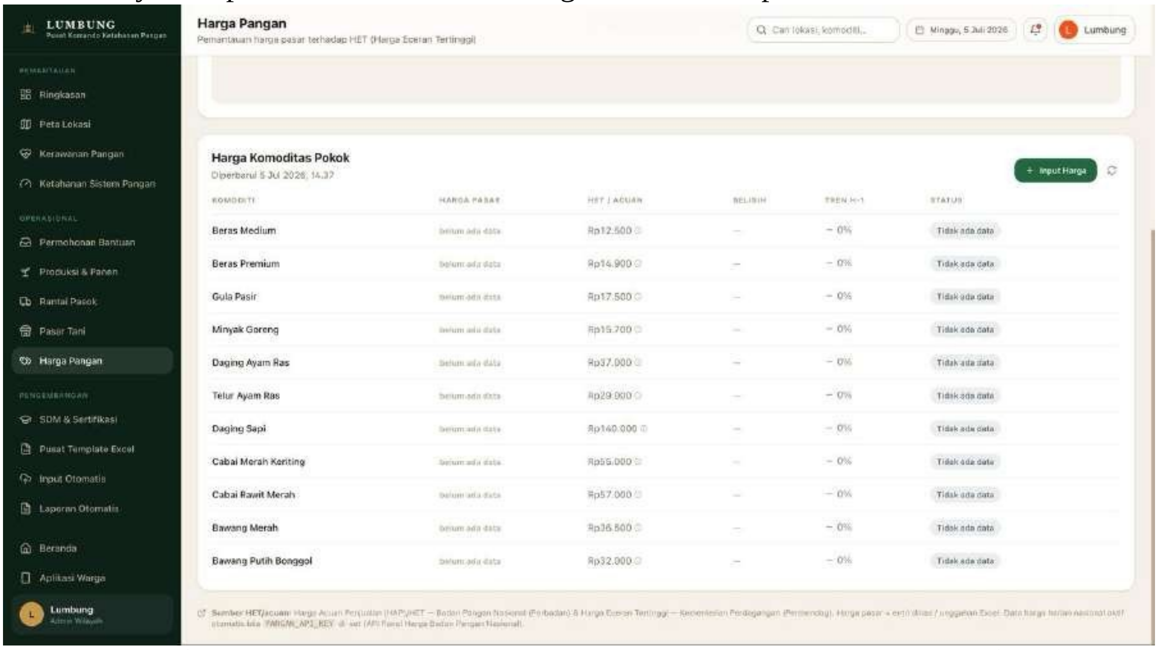


Figure 1. View of the Food Price monitoring dashboard on the LUMBUNG platform, showing a comparison of market prices against the Highest Retail Price (HET) for staple food commodities.
Source: LUMBUNG Prototype (https://lumbung-monitoring.vercel.app/), 2026.

Innovation Advantages, Local Products, and Intellectual Property (IP) Potential

We offer a smart monitoring and evaluation product based on technology and AI to produce a Food System Resilience Dashboard capable of monitoring, in real time, data on food insecurity, food security, and the social, economic, consumption, and environmental conditions of the region. Digital adoption is highly important because:

1) Rapid reach: smartphones have reached the majority of urban residents, and GPS and cameras have become low-cost, widely distributed food sensors
2) Spatial: GPS coordinates from residents can spatially map food conditions, something that would not be possible to do manually
3) Speed: real-time data allows the agency to see surges in assistance requests within minutes rather than weeks
4) Integration: a single centralized database ensures that residents and the agency view the same data without information silos

This AI-based technology is intended not only to display data but also to generate weekly periodic reports, directly calculate the region's food security and food insecurity levels, and recommend policies based on real-time data in the form of policy briefs. This innovative product is developed by utilizing local product components, namely GPS and CCTV already installed in various locations across the North Jakarta area.

We have named the Food System Resilience Dashboard for the urban farming community in the North Jakarta area

LUMBUNG. In this regard, LUMBUNG is a two-sided digital food security platform that connects residents (food reporters) and the government (monitors and decision-makers) within a single real-time, spatial- and AI-based data ecosystem.

The real-time data includes:

1) A map of household food insecurity per community in the North Jakarta area, and local market price data compared with urban farming output prices, representing the Accessibility dimension of the Food Security pillar

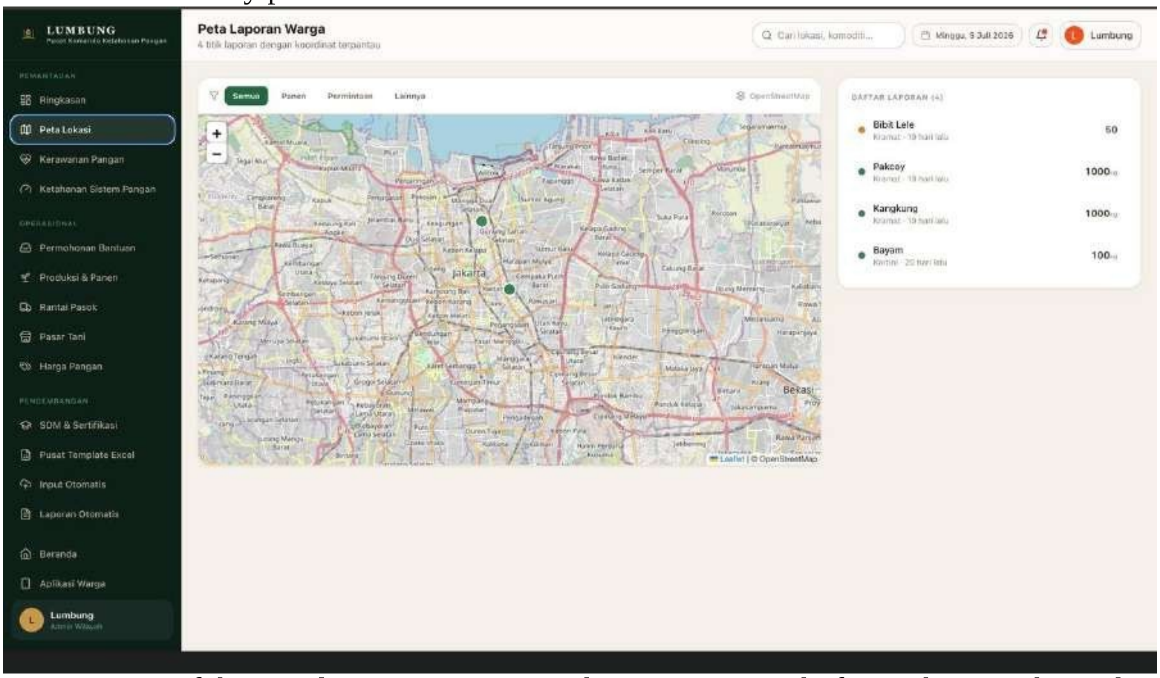
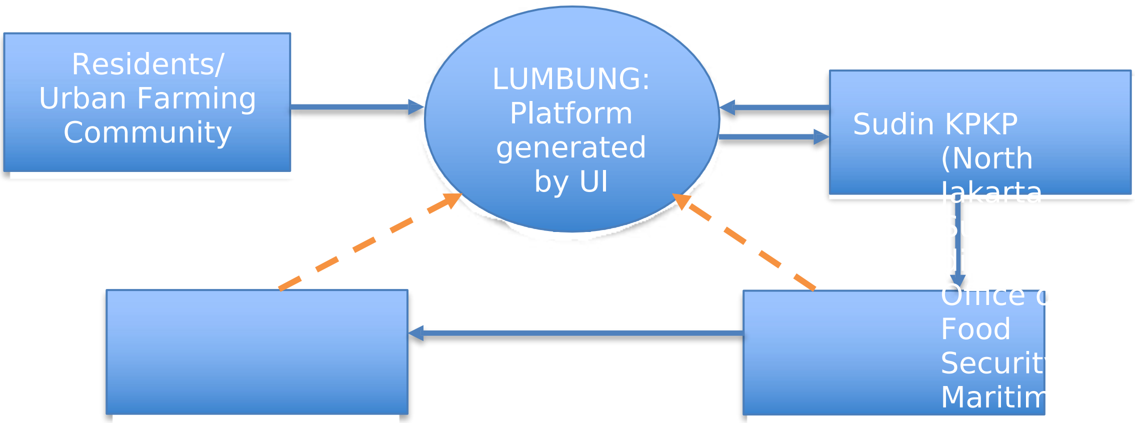


Figure 2. View of the Resident Report Map on the LUMBUNG platform, showing the real-time distribution of production and food-need report points from the urban farming community.
Source: LUMBUNG Prototype (https://lumbung-monitoring.vercel.app/), 2026.

2) Availability: community harvest data (kg/week), number of active plots of land, and crop diversification
3) Utilization: Consumption diversification (Household Dietary Diversity)
4) Stability: Variance/volatility of the 3 food security indicators over time

Figure 3. View of the LUMBUNG Smart Monitoring Food System Resilience conceptual model.
Source: Research team analysis, 2026.

This platform innovation, named LUMBUNG, has the potential to become intellectual property for Universitas Indonesia, since the platform also incorporates an intervention by Universitas Indonesia to take part in monitoring food security and food insecurity conditions in the North Jakarta area, which can later serve as a real-time data source for ongoing research purposes. Synergy among residents, government, and the university will strengthen food product innovation in the future.



Target and Market Share of the Food System Resilience Dashboard Product

User groups—namely the government and the public—who have different needs are brought together within a single ecosystem

Resident Targets:

1) Urban farming groups: can provide harvest information (type of food, quantity, and location of farmland), submit requests for emergency food assistance, and track ticket status
2) Fishers: provide information on fish catches and distribution needs for marine fishery products

3) PKK (Family Welfare Movement) cadres: monitor and report household food needs at the RT/RW (neighborhood unit) level
4) General public: view and compare food prices from the urban farming community against the HET price in the market

Government Targets:

1) Head of the KPKP Agency: information related to food security and food insecurity conditions in the North Jakarta area
2) Head of the Sudin KPKP North Jakarta: monitoring, evaluation, and periodic reports on food security and food insecurity by area in North Jakarta, as well as policy briefs
3) Agricultural extension officers: verify residents' harvest reports and coordinate urban farming groups in the field
4) Distribution officers or KPKP staff: manage the queue of assistance request tickets and carry out follow-up
5) Food data analysts: analyze production trends, price fluctuations, and monitor the regional food insecurity index and regional food security index
6) City Government/Mayor: receive periodic executive summaries of food security and food insecurity conditions

Note: Trademark to be registered; P3 does not require a patent; AI for outputs — scientific works, HKI (Intellectual Property Rights), patents (technology, product, method, etc.); network code or software technology features (improving transmission) must represent significant improvement.

C. TECHNOLOGY READINESS OF THE INNOVATION PRODUCT & METHODOLOGY

C.1. Roadmap for Achieving Innovation Downstreaming (maximum 3 pages)
Describe the stages of activity in the development of the product/technology innovation or the policy/model innovation that has been, is being, and will be carried out. The product development roadmap uses a timeframe (in years, e.g., 2024, 2025, 2026, etc.) as well as the TKT (Technology Readiness Level) status at each stage. The proposer may provide information on activities already carried out to explain the importance of the program's continuity through the proposed funding. The roadmap must illustrate the research progress from ***before*** *this year until it produces the final downstreaming output.*

Also provide an overview of the availability and continuity of raw materials/supply chain and production readiness (if relevant).

The description may be in narrative form accompanied by images and/or diagrams.

Roadmap for Achieving Innovation Downstreaming:

This downstreaming-based research began with research we previously conducted in 2025 in the Kampung Muara Bahari area, Tanjung Priok Sub-district, North Jakarta. Our research findings in this area, which is considered prone to narcotics, showed a high level of food insecurity among residents, with a moderate-to-severe category above 50%. Many factors contribute to this condition, one of which is accessibility to food, as indicated by the relatively low ability to purchase food in the market. Economic problems are one of the most dominant factors; therefore, government intervention is needed in the form of a food security program through an urban farming movement. The advantage of this program is that it reduces residents' dependence on food available in the market. Fluctuations in market food prices often place particular pressure on household expenditure, especially for those at the low-income level (poor and extremely poor). This program also serves as a solution for residents to engage in positive and productive activities rather than becoming involved in narcotics networks — even though the income may be substantial and immediate, the security and health risks for residents are very high. The worst-case risks include imprisonment, chronic illness, and even death. These risks must certainly be avoided through the contribution of all parties, including academics. Based on the results of this research, we developed a roadmap that is expected to be commercialized, with results that will benefit residents in North Jakarta.

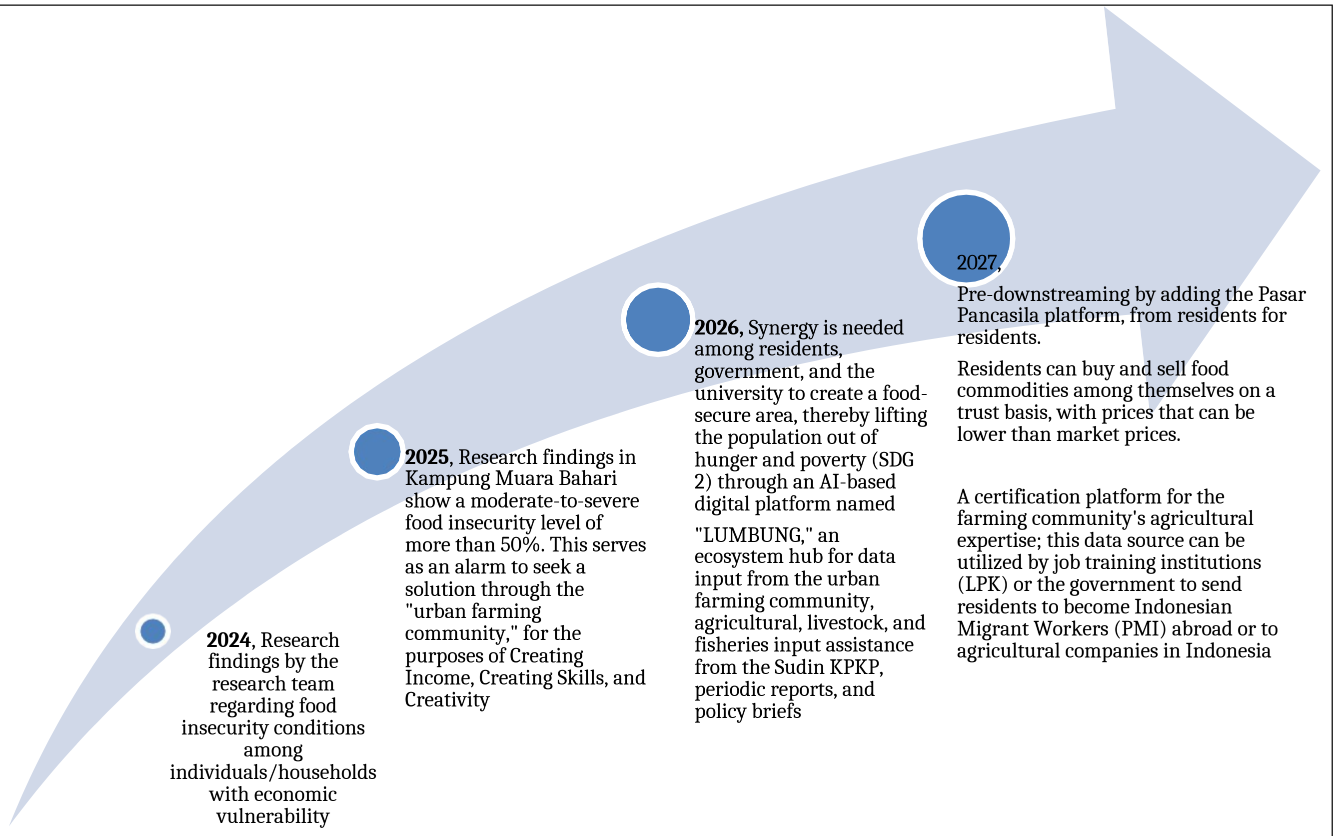


Figure 4. Downstreaming roadmap for the LUMBUNG Food System Resilience Dashboard product, 2024-2027.
Source: Research team analysis, 2026.

The downstreaming principle for the product, in the form of a digital application platform, will be developed by us for the benefit of stakeholders. Specifically, in this year 2026, the LUMBUNG platform is dedicated to non-commercial purposes, namely assisting local government in monitoring, evaluation, and accurate, precise, and rapid decision-making through the LUMBUNG channel to achieve SDG 2 (lifting residents out of hunger and poverty). Subsequently, once the platform is running well, we will expand into a commercial platform that benefits the private sector, residents, and the relevant government. In this regard, the information offered is:

1) a farming skills certification institution for urban farming community members interested in obtaining a license (managed by Universitas Indonesia)
2) Pasar Pancasila, from residents for residents, in the form of social media and a digital platform to facilitate the buying and selling of food commodities among residents, at prices that can be equal to or lower than general market prices. This platform upholds honesty and humanity, and therefore includes a food-sharing feature and charity food features that benefit poor and near-poor residents
3) Data on certified agricultural labor for users, namely agricultural companies abroad (Japan, Korea, the Netherlands) and domestic private companies (large domestic agricultural companies)

C.2. Program Implementation Method (maximum 4 pages)
Describe the method and approach for program implementation, including the scope of development together with partners, the activities involved, the outputs to be achieved from the activities carried out, and the role of each stakeholder (researcher/inventor and partner) in the program's implementation. The research method must be able to justify the proposed RAB (Budget Plan).

The description may be in narrative form accompanied by images or diagrams.

a. Program Implementation Approach and Method

The development program for the Food System Resilience dashboard in the North Jakarta area, named LUMBUNG, which will display various real data and food security and food insecurity conditions, will be implemented using two approaches, namely: 1) *Participatory Action Research* (PAR). This method is used to build a collaborative ecosystem among residents, government, and the university so that the development process is bottom-up, actively involving partners from needs identification and co-design of indicators through to program evaluation. The purpose of using the PAR method is to ensure the sustainability and sense of ownership of partners over the system being built. The second approach is 2) *Design Science Research Methodology* (DSRM), combined with the *Agile System Development Life Cycle (*SDLC) method. Both methods are used for the technical development of an AI-module-based dashboard in an iterative manner (Build-Measure-Learn), so that the system can later be adapted to the real needs of users in the field. This second approach is intended to ensure that the system can be tested, validated, and refined in a measurable way through clear stages.

b. Partnership Activity Approach through a Cooperation Agreement (PKS)

In the cooperation agreement between partners (government-residents and the university), the agreement serves as a guide for carrying out activities, including: **1) Data cluster:** development of an integrated database that combines primary data (agricultural, livestock, and fisheries harvest results reported by residents or village officials) and secondary data (food prices at Jakarta markets collected automatically using AI/web-scraping, as well as official government data sources). **2) System/dashboard cluster:** development of a web-based platform with three main modules (a resident reporting module for harvest input and assistance requests), a government response module for verification, assistance distribution, and mentoring, and a university analysis module for index processing, prediction, and visualization. **3) Artificial Intelligence (AI) cluster:** development of a module for extracting and cleaning food price data from online sources, detecting price anomalies as an indication of food insecurity, and a simple prediction model for short-term supply-demand projections. 4) **Index and policy cluster:** formulation and calibration of the Regional Food Security Index (IKPW), the Food Insecurity Index (IKP), and the Regional Food Access/Price Index, as well as an early warning mechanism for the government. 5) **Capacity-strengthening cluster:** education and training for residents (digital literacy and cultivation techniques) as well as for village officials/Sudin KPKP (data literacy and index interpretation for decision-making)

c. Activities and Outputs per Implementation Stage

Program implementation is divided into five stages over twelve (12) months, namely:

1) **Activity Stage 1:** : **Initiation and needs assessment (to be prepared within two months)**

Activity: an initial FGD (focus group discussion) was held with the Sudin KPKP, village representatives, and the urban farming community/urban farming-livestock-fishery groups to map data needs and services, field mapping — namely the location and number of urban farming communities, reference markets for price data, and the current bureaucratic flow for assistance requests — and the drafting and signing of a Cooperation Agreement (PKS) or a tripartite Memorandum of Understanding that regulates the division of roles and the data exchange mechanism. The output of this first stage is a needs assessment document for the ecosystem, a distribution map of the assisted urban farming communities and reference price markets, and a PKS document signed by the three parties.

The implementers in this first stage are the research team, the Sudin KPKP, and the village government, and the urban farming community

2) **Activity Stage 2: System Design and Measurement Indicators (month 2-4)**

The activities carried out in this second stage consist of: designing the system architecture and dashboard database structure; co-designing the definitions and formulas for the Regional Food Security Index, the Food Insecurity Index, and the Food Access/Price Index together with the Sudin KPKP so that they are aligned with the agency's operational definitions; designing the harvest reporting module, the assistance request module, and the government response module; and designing the architecture of the AI module for the automatic collection of food price data from markets in Jakarta

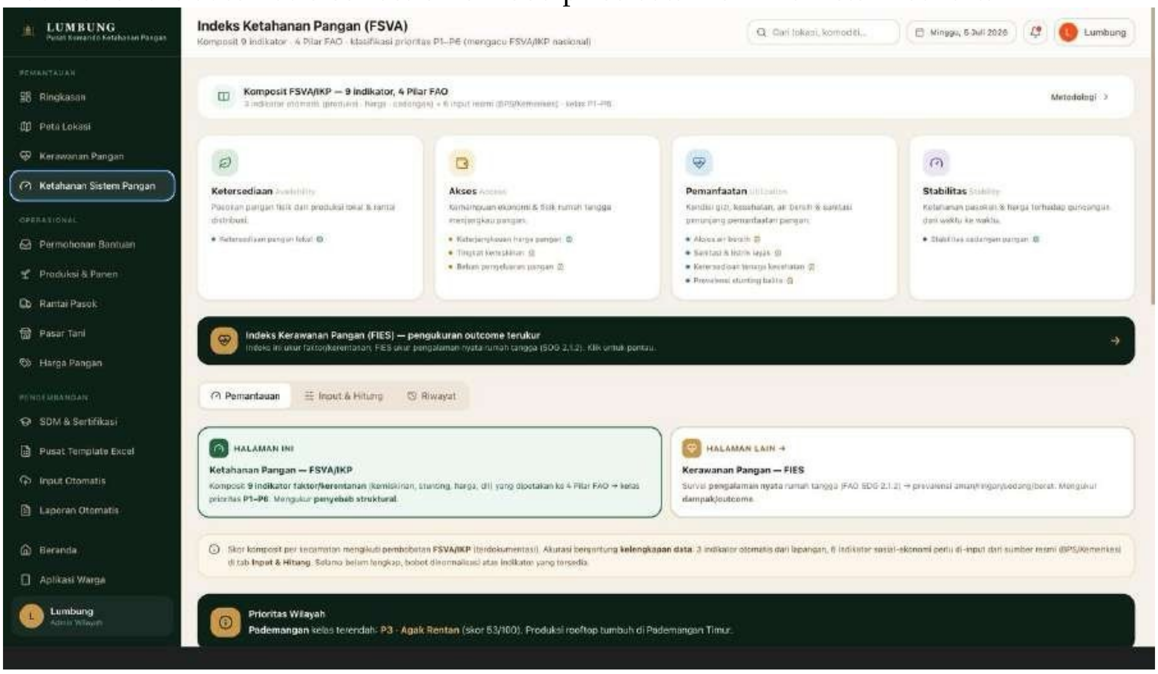


Figure 5. View of the Food Security Index (FSVA/IPJK) dashboard on the LUMBUNG platform, displaying a composite of 9 indicators based on the 4 FAO Pillars for regional priority classification (P1-P6).

Source: LUMBUNG Prototype (https://lumbung-monitoring.vercel.app/), 2026.

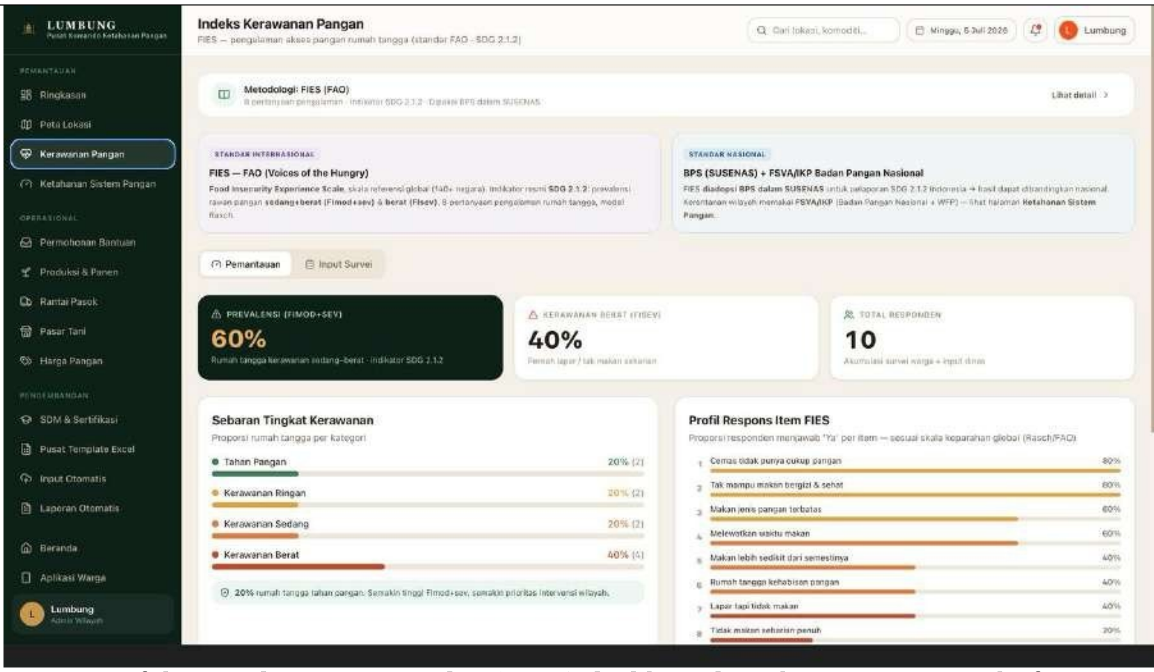


Figure 6. View of the Food Insecurity Index (FIES) dashboard on the LUMBUNG platform, measuring household food insecurity experience according to FAO standards (SDG 2.1.2).
Source: LUMBUNG Prototype (https://lumbung-monitoring.vercel.app/), 2026.

The implementers in this second stage include the research team, Sudin KPKP (North Jakarta Sub-district Office of Food Security, Maritime Affairs and Agriculture), and the urban farming community

3) Activity Stage 3: Development and Limited Trial (Pilot Project) (Months 4-7)

The activities in this third stage consist of iteratively developing the dashboard's back-end, front-end, and database; developing and training an AI model for food price data extraction and price anomaly detection; developing an automatic index calculation module; conducting a limited trial in several selected sub-districts (kelurahan) and communities; and improving the system based on feedback from the pilot results.

The main implementers consist of: the research team, the sub-districts (kelurahan) and urban farming communities selected as pilot project sites, and Sudin KPKP

4) Activity Stage 4: Implementation, Training, and Mentoring (Months 7-10)

The activities in this fourth stage consist of outreach and training on the use of the dashboard for residents/urban farming communities, covering harvest data input and the submission of applications for agricultural, livestock, and fisheries assistance. Training is also provided for sub-district (kelurahan)/Sudin KPKP officials on data verification, rapid response, and interpretation of theindexin decision-making. The next activity is the full launch of the dashboard for the North Jakarta region, along with periodic field mentoring provided by the government and research team to urban farming communities receiving assistance

The main implementers consist of: the research team, the sub-districts (kelurahan) and urban farming communities selected as pilot project sites, and Sudin KPKP

5) **Activity Stage 5: Monitoring, Evaluation, and Policy Dissemination (Months 10-12)**

The activities in this fifth stage consist of periodic monitoring of system usage and data quality, and analysis of trends in the IKPW, IKP, and Food Access Index by sub-district (kelurahan) for the preparation of a*Policy Brief*.Other activities include an FGD (Focus Group Discussion) to validate the results and prepare a sustainability plan together with partners, as well as dissemination of results through a*Policy Brief*and scientific publications.

The main implementers consist of: the research team, the urban farming communities selected as pilot project sites, and Sudin KPKP

In this regard, we have developed the Smart Monitoring Food System Resilience Dashboard website.https://lumbung-monitoring.vercel.app/

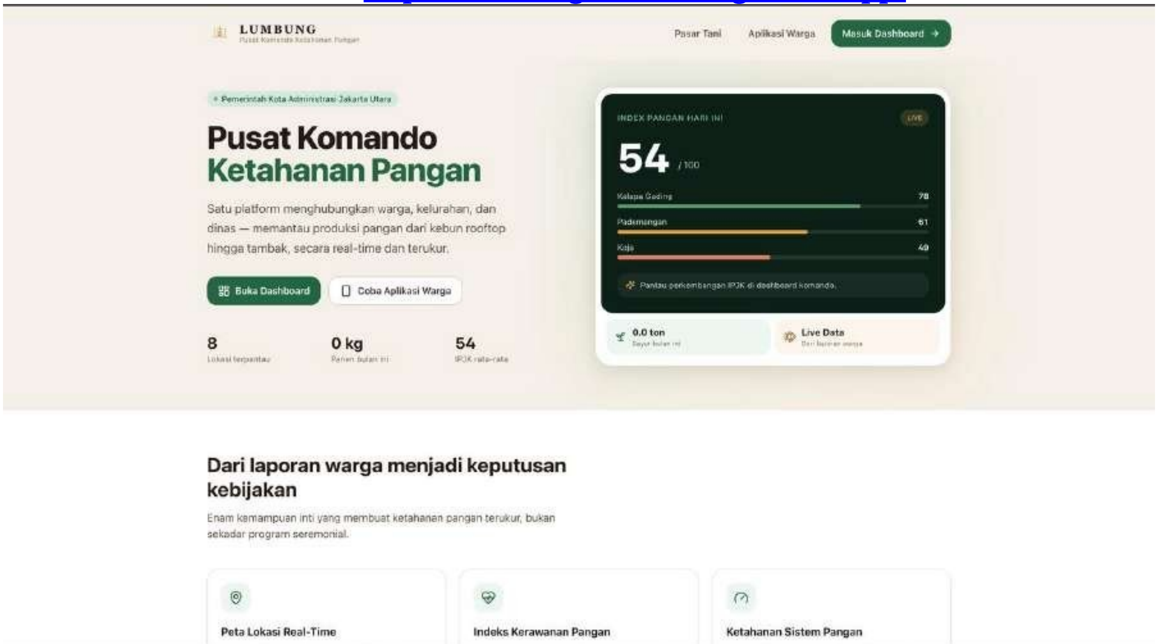


Figure 7. Main page (landing page) view of the LUMBUNG platform, displaying the daily Jakarta Utara Food Index (IPJK) and a summary of regional monitoring.
Source: LUMBUNG Prototype (https://lumbung-monitoring.vercel.app/), 2026.

Role of Stakeholders

The success of the collaborative ecosystem depends on a clear division of roles among researchers/Universitas Indonesia, government, and residents/community, as summarized in the following table.

Stakeholder	Role and Form of Contribution
Researchers / University Team *System Developer, Data Processor, and Mediator*	• Designing and developing the dashboard architecture and AI module • Processing real-time data into the Regional Food Security Index, Food Vulnerability Index, and Food Access Index • Preparing periodic policy briefs as data-driven recommendations for the government • Providing education and data/digital literacy training for residents and government officials alike • Acting as a mediator bridging the interests of residents and government so that policies are responsive as well as evidence-based

	• Maintaining methodological quality: index validity, data security, and ethics of AI use • Conducting testing of the dashboard (prototype) in specific areas
Government (North Jakarta Sudin KPKP & Sub-district/Kelurahan Offices) ***Verifier, Service Provider, and Policy Maker***	• Verifying harvest data and assistance applications submitted by residents/communities • Providing and distributing agricultural, livestock, and fisheries assistance (seeds, fertilizer, technical training) based on data in the dashboard • Providing a rapid response to regional food vulnerability conditions detected by the system • Providing technical and institutional mentoring to communities receiving assistance • Following up on policy briefs from the university in formulating regional food security policy • Serving as a source of administrative data and providing institutional legitimacy for the program
Residents / Community (Urban Farming, Urban Farming-Livestock-Fisheries Groups) ***Primary Data Provider and Beneficiary***	• Inputting harvest data periodically through the dashboard, either independently or through sub-district (kelurahan) officials • Submitting applications for production input assistance (seeds, fertilizer) or training as needed • Reporting on the utilization of assistance received as a form of accountability • Actively participating in FGDs, pilot trials, and training • Providing feedback on the ease of use of the system (usability feedback)

The three roles described above (Government (Sudin KPKP), Universitas Indonesia, and Residents/Urban Farming Community) form a sustainable work cycle: residents report their harvest results, the system processes the data together with food price data into various index measurements, the university then formulates indices and recommendations, the government then responds through assistance and policy, and residents in turn report on the utilization of the assistance received. This Triple Helix synergy aims to improve regional food security, maintain residents' welfare, and further strengthen residents' trust in the government.

d. Budget Plan Aligned with Activity Stages

Stage	Main Cost Component	Justification of Need
Stage 1 Initiation & Needs Assessment	Honoraria for FGD resource persons/moderators; meeting refreshments; transportation & daily allowances for field surveys; stationery (ATK) and documentation	The cost requirements for FGDs and field surveys are proportional to the number of sub-districts (kelurahan) and communities that must be mapped across the North Jakarta region, so the amount can be calculated based on the number of locations and participants involved.
Stage 2 System & Indicator Design	Honoraria for information systems experts/data scientists; co-design workshop costs for indicators; design software licenses/subscriptions	Formulating the index and system architecture requires specific technical expertise (data science, system design) that is generally not fully available within the core team, so the expert honoraria are justified by the complexity of designing the multi-indicator index and AI module.
Stage 3 Development & Pilot Trial	Honoraria for programmers/developers; server rental/cloud computing and AI API	This component represents the largest portion of the RAB (Budget Plan) because the system operates in real time and the AI module requires

	costs; mobile credit/internet data costs; implementation costs for field trials	computing capacity and ongoing paid API access during the development and pilot period (approximately 4 months); developer honoraria are calculated proportionally to the duration of the development sprints.
Stage 4 Implementation, Training & Mentoring	Honoraria for training facilitators; printing costs for modules/materials; refreshments and venue rental for training; launch/outreach costs; transportation costs for field mentoring	The number of training participants (residents and officials across a number of sub-districts/kelurahan) forms the basis for calculating the requirements for refreshments, module reproduction, and facilitator honoraria per training session, so that the budget is measurable and commensurate with the coverage of the target area.
Stage 5 Monitoring, Evaluation & Dissemination	Honoraria for monitoring enumerators; validation FGD costs; policy brief preparation and printing costs; dissemination publication/seminar costs; final reporting costs	The budget is commensurate with the program's mandatory output targets (policy brief and scientific publications) in accordance with the grant scheme, as well as the need to validate results together with partners before the program results are widely disseminated.
Program Management (all stages)	Stationery (ATK) and team communication; coordination and internal meetings; administrative monitoring	Allocated within a reasonable limit (generally 10-15% of the total budget in accordance with the grant scheme guidelines) to ensure smooth cross-party coordination throughout the program cycle without burdening other substantive components.

C.3. Outputs and Achievement Targets (in accordance with the guidelines)				
No.	**Output**	**Achievement Target**		**Related Activity**
		Quantity	**Status**	
MAIN OUTPUTS				
1	*TKT (Technology Readiness Level) 4 prototype*	*1*	*completed*	*North Jakarta Region Food System Resilience Dashboard*
2	*product testing document*	*1*	*completed*	Index and methodology document (formulas and documentation of the validated IKPW, IKP, and Food Access Index)
3	*HKI (Intellectual Property Rights) in the form of copyright registered under UI's name*	*1*	*completed*	Registration of copyright for the computer program, database structure, and module index calculation
4	*Policy Brief*	*1*	*Completed*	Policy recommendations submitted to and followed up on by Sudin
5	*Publication in a Reputable International Journal (Scopus Index Q1 and Q2)*	*1*	*Submitted*	
ADDITIONAL OUTPUTS				
1	*scientific publication*	*1*	*Submitted*	Article in a Reputable International Scientific Journal
2	Copyright-protected training module	1	Completed	Training module or book
3	Popular article in the mass media	1	Completed	News item for public dissemination
4	Program documentation video	1	Completed	Video uploaded to YouTube

C.4. Resources Required Based on the Program Implementation Method			
Activity Name	**Component**	**Amount of funding and its source (thousand rupiah)**	
		UI	**Partner (In kind)**
Output	*TKT 7 Prototype — North Jakarta Food System Resilience Dashboard (LUMBUNG)*		
Activity Group	*Testing & validation of the index, limited trial (pilot project) in selected sub-districts/communities*		
1. System Initiation & Design (Phase 1–2)	*Needs-assessment FGD, tripartite PKS, system architecture design & IKPW/IKP/Food Access Index indicators*	Rp36.600.000	Rp0
2. Development & Pilot Testing (Phase 3)	Development of system modules (Next.js, Supabase, AI integration) and implementation of limited testing (pilot project) in selected urban farming sub-districts/communities	Rp2.500.000	Rp0
3. Implementation, Monitoring, Evaluation & Dissemination (Phase 4–5)	Training & mentoring for residents/officials, dashboard launch, monitoring-evaluation, policy brief preparation, dissemination (scientific publication), IT & AI infrastructure management, program management, and HKI (Intellectual Property Rights) recording throughout the 12-month program	Rp160.900.000	Rp0
	Subtotal	**Rp200.000.000**	**Rp0**
	Grand Total	**UI: Rp200.000.000 | Partner (in-kind): Rp0**	

C.4.1. Detailed Budget Plan (RAB)

The following RAB details are prepared according to the 5 activity stages and the IT & AI Infrastructure component, with a Grand Total of exactly Rp200.000.000 in line with the funds proposed to UI.

Phase	No.	Component Description	Volume	Unit	Unit Price (Rp)	Amount (Rp)
PHASE 1: Initiation & Needs Assessment (Month 1-2)						
	1	Honorarium for speakers/moderators of the initial FGD with Sudin KPKP (North Jakarta Sub-district Office of Food Security, Maritime Affairs and Agriculture), sub-districts & communities	1	package	Rp1.500.000	Rp1.500.000
	2	Meeting/FGD refreshments	1	package	Rp1.500.000	Rp1.500.000
	3	Transportation & daily allowance for field survey (6 districts)	1	package	Rp2.000.000	Rp2.000.000
	4	Office supplies & documentation	1	package	Rp1.000.000	Rp1.000.000
					Subtotal PHASE 1	**Rp6.000.000**
PHASE 2: System & Indicator Design (Month 2-4)						
	1	Honorarium for information systems expert/data scientist (formulation of IPJK, FIES) - 2 months	2	month	Rp3.000.000	Rp6.000.000
	2	Honorarium for programmer/developer (Next.js, Supabase, AI module) - 4 months	4	month	Rp5.400.000	Rp21.600.000
	3	Indicator co-design workshop with Sudin KPKP	1	package	Rp2.000.000	Rp2.000.000
	4	Design tool license/subscription (UI/UX, wireframe)	1	package	Rp1.000.000	Rp1.000.000
					Subtotal STAGE 2	**Rp30.600.000**
STAGE 3: Development & Pilot Testing (Month 4-7)						
	1	Cost of conducting the limited trial/pilot project	1	package	Rp2.500.000	Rp2.500.000
					Subtotal STAGE 3	**Rp2.500.000**
STAGE 4: Implementation, Training & Mentoring (Month 7-10)						
	1	Honorarium for facilitators training residents & officials	1	package	Rp2.000.000	Rp2.000.000
	2	Printing of training modules/materials	1	package	Rp1.500.000	Rp1.500.000
	3	Refreshments & training venue rental	1	package	Rp3.500.000	Rp3.500.000
	4	Cost of launching/promoting the LUMBUNG dashboard	1	package	Rp1.500.000	Rp1.500.000
	5	Transportation for periodic field mentoring	1	package	Rp2.500.000	Rp2.500.000
					Subtotal STAGE 4	**Rp11.000.000**
STAGE 5: Monitoring, Evaluation & Dissemination (Month 10-12)						
	1	Honorarium for monitoring enumerators (2 persons x 3 months)	6	PT	Rp600.000	Rp3.600.000
	2	Cost of FGD (Focus Group Discussion) to validate results with partners	1	package	Rp2.000.000	Rp2.000.000
	3	Preparation & printing of Policy Brief	1	package	Rp1.000.000	Rp1.000.000
	4	Scientific publication (international journal, Scopus Q1/Q2) & dissemination seminar	1	package	Rp7.000.000	Rp7.000.000
	5	Final reporting (reproduction, binding, documentation)	1	package	Rp1.000.000	Rp1.000.000
					Subtotal STAGE 5	**Rp14.600.000**
INTELLECTUAL PROPERTY RIGHTS (HKI)						
	1	Cost of copyright registration in the name of Universitas Indonesia (computer program, database, index module)	1	package	Rp400.000	Rp400.000
					Subtotal INTELLECTUAL PROPERTY RIGHTS	**Rp400.000**

PROGRAM MANAGEMENT (All Stages, 12 Months)						
	1	Office supplies & team communication	1	package	Rp1.400.000	Rp1.400.000

Stage	No.	Component Description	Volume	Unit	Unit Price (Rp)	Amount (Rp)
	2	Team coordination & internal meetings	1	package	Rp2.000.000	Rp2.000.000
	3	Administrative monitoring & evaluation of the program	1	package	Rp1.500.000	Rp1.500.000
					Subtotal PROGRAM MANAGEMENT	**Rp4.900.000**
IT & AI INFRASTRUCTURE (12 months)						
	1	Cloud & Hosting Infrastructure (dashboard hosting, database, domain+SSL)	1	year	Rp10.166.000	Rp10.166.000
	2	Database, Storage & Backup (automatic backup & disaster recovery)	1	year	Rp3.222.000	Rp3.222.000
	3	System Security & Monitoring (WAF, uptime monitoring, error tracking)	1	year	Rp17.184.000	Rp17.184.000
	4	User Management & Productivity (version control, product analytics, transactional email)	1	year	Rp14.058.000	Rp14.058.000
	5	GIS & Mapping Integration (geocoding, interactive map of report locations)	1	year	Rp5.370.000	Rp5.370.000
	6	AI Infrastructure for system development (coding assistant, technical documentation, testing)	1	year	Rp40.000.000	Rp40.000.000
	7	AI Infrastructure for platform production features (automated reporting, dashboard summaries, data analysis & recommendations)	1	year	Rp40.000.000	Rp40.000.000
					Subtotal IT & AI INFRASTRUCTURE	**Rp130.000.000**
GRAND TOTAL RAB LUMBUNG (UI Funds)						**Rp200.000.000**

C.5. Activity Schedule													
Activity/Task Name	**2026**						**2027**						
	Jul	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**
1. Activity 1													
a. : Initiation and needs assessment													
2. Activity 2: System design and measurement indicators													
3. Activity 3: Development and limited trial (pilot project)													
4. Activity 4: Implementation, Training and Mentoring													
5. Activity 5: Monitoring, evaluation and policy dissemination													

D. PARTNER CAPACITY AND COMMITMENT *(maximum 1 page)*

Explanation of the partner, its relevance to the proposed innovation, its track record of collaboration with the lecturer/inventor, and its contribution. Describe the partner's suitability to the innovation to be developed, as well as the partner's role in the innovation development stage through to downstreaming. This should also include the form of the partner's contribution to the development of the proposed program's products/outputs.

- *For* ***DUDI (business/industry world) partners,*** *it is necessary to explain the relationship between the core business and the relevance of the science, technology, and health innovation product/technology to the partner's core business, as well as plans for further development.*
- *For* ***Government Institution partners,*** *it is necessary to explain the need for and urgency of the social innovation policy/model being designed, as well as its alignment with the government's roadmap, and the commitment to adopt the results of the innovation.*

- **Partner:** Sudin KPKP (North Jakarta Sub-district Office of Food Security, Maritime Affairs and Agriculture) — a regional government institution overseeing the region's food security program, including the development of urban farming communities, the distribution of agricultural/livestock/fisheries assistance, and city food policy.
- **Relevance:** The need for real-time monitoring & evaluation, which is at the core of LUMBUNG, is a daily operational need of Sudin KPKP that has not been met to date due to the manual system in use.
- **Track record:** Collaboration has been established since preliminary research in 2025 in Kampung Muara Bahari, Tanjung Priok; it was reaffirmed through coordination with the Head of Sudin KPKP North Jakarta on 12 June 2026.
- **Partner's role per stage:** FGD & signing of the PKS (initiation) → operational definition of the index (system design) → provision of field access (development & pilot) → data verification and follow-up on assistance requests (implementation) → follow-up on the policy brief (monitoring-evaluation).
- **Partner's contribution:** provision of administrative data and institutional legitimacy, field verification of citizen reports, distribution of production assistance based on dashboard data, and a commitment to adopt LUMBUNG as the department's routine working system after the funding period ends.

E. PRODUCT DOWNSTREAMING READINESS *(maximum 1 page)*

This section must be completed by the team together with the partner. Provide an explanation of market needs (and market testing if already conducted), the pricing and cost plan, business feasibility, investment needs/feasibility, as well as the commercialization strategy. Include a screenshot of the Business Model Canvas and provide a descriptive explanation.

Market Needs:

Market need for the Food System Resilience Dashboard (LUMBUNG) comes from two sides: the government (Sudin KPKP North Jakarta), which requires a real-time monitoring and evaluation system to replace the manual mechanism that has been in use to date, and the urban farming/fishing communities, which need an official channel to report production results and submit requests for assistance. Formal market testing has not yet been conducted because the platform is still at the non-commercial stage (TKT 7, heading toward full use by the government partner); validation of user needs was carried out through an initial FGD with Sudin KPKP, the kelurahan (sub-district office), and the urban farming community during the program's initiation stage. **Pricing and Cost Plan:**

In 2026, LUMBUNG will be operated on a non-commercial basis as a local government support tool, and therefore no charge will be imposed on either the government partner or citizen users. The

operational cost requirements (hosting, database, system security, and AI infrastructure) refer to the program's Budget Plan (RAB), with IT & AI infrastructure needs amounting to Rp130.000.000 per year out of total program funds of Rp200.000.000 (details in the RAB attachment). The pricing model for the commercial phase (farming skills certification, a food marketplace among residents, and certified agricultural labor data) will be determined together with the partner once the non-commercial platform has proven

to operate stably.

Business Feasibility and Investment Needs:

Short-term business feasibility (2026) is measured by the sustainability of adoption by the government partner and the level of participation of the urban farming community, rather than by commercial margin. For the commercial phase (2027 onward), the main investment needs include increasing server/cloud capacity for a larger user scale, developing the marketplace and certification modules, as well as collaboration with distribution partners/overseas agricultural labor partners. A detailed investment feasibility study will be prepared together with the partner during the first-year-end evaluation of the program.

Commercialization Strategy:

The downstreaming strategy follows the two-stage roadmap described in section C.1: (1) in 2026, LUMBUNG will operate on a non-commercial basis to support SDG 2 in the North Jakarta area — at this stage, the module for marketing residents' harvest produce ("Pasar Tani") is already actively running on a non-commercial basis within the LUMBUNG dashboard, forming the foundation for subsequent commercial development; (2) after 2026, LUMBUNG will be developed into a commercial platform with three lines: a farming skills certification institution (managed by UI), a wider-scale food marketplace among residents ("Pasar Pancasila") with sharing/charity food features, and a database of certified agricultural laborers for both domestic and overseas corporate users

Business Model Canvas — LUMBUNG

The following Business Model Canvas summarizes LUMBUNG's business model from the non-commercial stage (2026, funded by the UI Innovation Funding Program, P3 Scheme) toward the commercial phase post-2026, in line with the RAB (Budget Plan) and the downstream commercialization roadmap in Sections C.1 and E.

KEY PARTNERS

- Sudin KPKP (North Jakarta Sub-district Office of Food Security, Maritime Affairs and Agriculture) (main government partner)
- Universitas Indonesia (research & HKI (Intellectual Property Rights))
- Sub-districts (Kelurahan) & urban farming/crop-livestock-fishery communities
- Cloud & AI providers (Vercel, Supabase, Anthropic)
- Agricultural labor distribution partners (commercial phase: Japan, Korea, Netherlands)

KEY ACTIVITIES

- Development & maintenance of the dashboard and the IPJK/FIES indices
- Field data collection & validation (FGDs, surveys, verification)
- Training & mentoring of residents/officials; monitoring & evaluation
- Development of commercial modules post-2026

KEY RESOURCES

- Research & development team (data scientists, developers, AI engineers)
- Cloud/AI infrastructure (Next.js, Supabase, Claude, Vercel)
- Real-time operational data from North Jakarta's 6 districts & HKI
- A live TKT (Technology Readiness Level) 7 prototype

VALUE PROPOSITIONS

- Real-time, AI-based food security monitoring & evaluation (IPJK & FIES), replacing manual processes
- Official channel for residents to report production & apply for assistance transparently
- Data-driven policy dashboard for Sudin KPKP (early detection of food insecurity)
- Pasar Tani (Farmers' Market): connects residents' harvests directly to buyers (already live)

CUSTOMER RELATIONSHIPS

- Direct & periodic mentoring by researchers and sub-district officials
- Data verification training for agency officers
- Technical support via the resident portal (self-service)

CHANNELS

- Web dashboard (resident portal & agency monitoring)
- FGDs, outreach, and in-person training
- Sub-districts (Kelurahan) & urban farming communities
- (Commercial phase) social media & distribution partners

CUSTOMER SEGMENTS

- Sudin KPKP (primary user/policymaker)
- Urban farming & crop-livestock-fishery communities (6 districts)
- Residents receiving food assistance
- (Commercial phase) local food buyers, industry partners/agricultural labor

COST STRUCTURE

- Stages 1-2 (initiation, system design, honoraria for experts & developers): Rp36.600.000
- Stages 3-5 (piloting, implementation, training, monitoring & dissemination): Rp28.100.000
- IT & AI infrastructure (cloud, security, GIS, AI - 12 months): Rp130.000.000
- Program Management & HKI: Rp5.300.000
- TOTAL: Rp200.000.000 (per the RAB)

REVENUE STREAMS

- 2026: non-commercial, fully funded by the UI Innovation Funding Program, P3 Scheme (Rp200.000.000)
- Post-2026 (commercial): farming skills certification licensing
- Pasar Pancasila marketplace transaction commissions
- Certified agricultural labor data (domestic & overseas partners)

List of Attachments:

1. Document *of self-assessment* for the current innovation TKT (Technology Readiness Level), along with supporting documents as evidence
2. Budget Plan (RAB) (only funds proposed to DIRBT UI, excluding partner contribution funds)
3. Proposer Profile
4. Researcher's Statement Letter by the Principal Proposer
5. Partner Profile
6. Partner's Statement of Willingness Letter
7. *Business Model Canvas* designed by the proposer and partner(s) for the commercialization strategy
8. Intellectual Property Certificates of the research team relevant to the topic of the proposal submitted

F. ATTACHMENT: INTELLECTUAL PROPERTY RIGHTS (HKI) CERTIFICATE

Attached below are the Certificates of Copyright Recordation (Hak Cipta) of the research team relevant to the topic of the proposal submitted.

1. **Certificate of Copyright Recordation - "Sustainable Agriculture-Based Food Security Strategy Adopting Society 5.0 in IKN: A Pancasila Economic Perspective (Economic-Spiritual)"**

REPUBLIK INDONESIA
KEMENTERIAN HUKUM

SURAT PENCATATAN CIPTAAN

Dalam rangka pelindungan ciptaan di bidang ilmu pengetahuan, seni dan sastra berdasarkan Undang-Undang Nomor 28 Tahun 2014 tentang Hak Cipta, dengan ini menerangkan:

Nomor dan tanggal permohonan	:	EC002026114301, 14 Juli 2026
Pencipta		
Nama	:	**Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA**
Alamat	:	Jl. Irida Barat IV D30-3, RT012/RW014, Kel. Bekasi Jaya, Bekasi Timur, Kota Bekasi, Jawa Barat, 17112
Kewarganegaraan	:	Indonesia
Pemegang Hak Cipta		
Nama	:	**Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA**
Alamat	:	Jl. Irida Barat IV D30-3, RT012/RW014, Kel. Bekasi Jaya, Bekasi Timur, Kota Bekasi, Jawa Barat, 17112
Kewarganegaraan	:	Indonesia
Jenis Ciptaan	:	**Karya Tulis Lainnya**
Judul Ciptaan	:	**Strategi Ketahanan Pangan Berbasis Pertanian Berkelanjutan Adopsi Society 5.0 di IKN: Perspektif Ekonomi Pancasila (Ekonomi-Spiritual)**
Tanggal dan tempat diumumkan untuk pertama kali di wilayah Indonesia atau di luar wilayah Indonesia	:	10 Desember 2024, di Kota Depok
Jangka waktu pelindungan	:	Berlaku selama hidup Pencipta dan terus berlangsung selama 70 (tujuh puluh) tahun setelah Pencipta meninggal dunia, terhitung mulai tanggal 1 Januari tahun berikutnya.
Nomor Pencatatan	:	001347491

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a.n. MENTERI HUKUM
DIREKTUR JENDERAL KEKAYAAN INTELEKTUAL
u.b
Direktur Hak Cipta dan Desain Industri

Agung Damarsasongko,SH.,MH.
NIP. 196912261994031001

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2. Certificate of Copyright Recordation - "How to Build Sustainable Food System Resilience Based on Society 5.0: Foundations of Pancasila Economics"

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Dalam rangka pelindungan ciptaan di bidang ilmu pengetahuan, seni dan sastra berdasarkan Undang-Undang Nomor 28 Tahun 2014 tentang Hak Cipta, dengan ini menerangkan:

Nomor dan tanggal permohonan	:	EC002026114309, 14 Juli 2026
Pencipta		
Nama	:	**Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA**
Alamat	:	Jl. Irida Barat IV D30-3, RT012/RW014, Kel. Bekasi Jaya, Bekasi Timur, Kota Bekasi, Jawa Barat, 17112
Kewarganegaraan	:	Indonesia
Pemegang Hak Cipta		
Nama	:	**Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA**
Alamat	:	Jl. Irida Barat IV D30-3, RT012/RW014, Kel. Bekasi Jaya, Bekasi Timur, Kota Bekasi, Jawa Barat, 17112
Kewarganegaraan	:	Indonesia
Jenis Ciptaan	:	**Karya Tulis Lainnya**
Judul Ciptaan	:	**Bagaimana Membangun Ketahanan Sistem Pangan Berkelanjutan Berbasis Society 5.0: Landasan Ekonomi Pancasila**
Tanggal dan tempat diumumkan untuk pertama kali di wilayah Indonesia atau di luar wilayah Indonesia	:	10 Desember 2024, di Kota Depok
Jangka waktu pelindungan	:	Berlaku selama hidup Pencipta dan terus berlangsung selama 70 (tujuh puluh) tahun setelah Pencipta meninggal dunia, terhitung mulai tanggal 1 Januari tahun berikutnya.
Nomor Pencatatan	:	001347505

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